Pick a Platter

Today, the entry-level drives are the 5400 rpm drives, which deliver sequential read and write speeds of about 25 MBps. This is sufficient for applications such as word processors, Internet surfing, etc. These drives are at least ATA/66 compliant, which implies a peak theoretical transfer rate of 66 MBps.

The higher performance 7200 rpm IDE models are generally ATA/100 compliant and are ideally suited to applications such as audio and video processing. The prices for these drives begin from Rs 4,000 to 4,300 for a 40 GB unit. The brand-new Serial ATA (SATA) drives support the ATA/150 standard. SATA II in fact even goes as far as enabling a theoretical throughput of 300 MBps! An 80 GB SATA drive costs around Rs 10,000.

When it comes to sheer performance, however, the latest SCSI technology still takes the honours with a theoretical transfer rate of 320 MBps, and the ability to daisy chain several devices on a single controller. However, this technology comes at a high price.

Smart Tips

Capacity: For most users an entry-level 40 GB drive would be sufficient. However, IDE drives of capacity up to 250 GB are available. SCSI models are available in sizes ranging from 9 GB to 80 GB.

Rotational speed: This refers to the spindle rpm of the drive under consideration. Higher spindle speeds lead to faster read and write performance. These range from 5400 to 7200 rpm for IDE drives and 7200 to 15000 rpm for SCSI models.

Sustained data transfer rates: This is a measure of the drive's performance in terms of data transfer to and from the drive. This should be about 20 to 30 MBps for 5400 rpm drives and 30 to 50 MBps for 7200 rpm drives. The new SATA drives support even higher rates.

Average seek time: This is the average time taken by the drive to access data stored on the drive. A time between 5.5 to 10 milliseconds is acceptable for most mainstream applications.

Data buffer (cache): This is the amount of onboard memory that the drive possesses. It increases transfer speeds when data is copied from one location in the drive to another. This ranges from 512 KB in the 5400 rpm drives to as much as 8 MB in the 7200 rpm IDE models. The most common data buffers size is 2 MB.

Motherboard/IDE controller card: In order to tap the full potential of a hard drive, you need to make sure that your motherboard's IDE controller supports the same specification (ATA/66/100/133) interface. If you're buying a SATA hard disk, your motherboard should have onboard SATA support, else you'll have to buy a controller card as well. Finally, be aware that there are two types of SATA controllers—native and bridge. A bridge merely acts as a connection between a SATA drive and the motherboard's standard parallel ATA interface, while a native controller is a true SATA controller. Since a bridge uses the same underlying technology that runs the current ATA interface, it wipes out SATA's performance advantage.

Decision Maker

	General home use	Focus on capacity	Cost no bar
You Need	A hard disk for general applications, such as word processing and Internet surfing	Large data storage capacity and speed is not such an issue.	You want the maximum capacity available. Speed becomes a top concern as well. Price is not a problem.
Look For	40 GB, 7200 rpm, supporting ATA/100 transfer rates.	A minimum of 80 GB, 7200 rpm, supporting at least ATA/100.	A 200 or 250 GB drive at 5400 rpm or 7200 rpm, supporting at least ATA/133.
Our Pick	Samsung SP4002H	Seagate Barracuda ATA IV 80 GB, Western Digital WD1200.	Maxtor DiamondMax 16 250 GB
Price Range	Up to Rs 4,500	Up to Rs 6,500	Up to Rs 25,000